REDUCTION OF THE VANISHING OF L-POWER TORSION COHOMOLOGY TO THE MAXIMAL PRO-L EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the equivalence, in both directions, and the two paragraphs after the set-up correspond exactly to those two directions.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Maximal pro-l quotient of the absolute Galois group and the corresponding extension. Same chapter, the maximal pro-l subgroup H of G = Gal(K^sep/K) and the field L of the corresponding extension. Supplies the field L and the group whose cohomology is compared with that of G.
- Comparison of Galois cohomology with etale cohomology of the spectrum of a field. Same chapter, the identification of H^n_etale(Spec(K), F) with the Galois cohomology H^n(G, F_sbar) for an l-power torsion sheaf. Supplies the translation used in both directions of the proof.
- Spectral sequence or inflation-restriction argument for a closed subgroup of finite index. Same chapter, the behaviour of Galois cohomology under the passage to a pro-l subgroup and the vanishing of the cohomology of an l-group with l-power torsion coefficients. Supplies the argument that vanishing on the subgroup implies vanishing for the whole group in the coefficients considered.

Source reference. Original lemma 0DV7: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 10543–10543. Navigation pointer invoked at proof line 10613; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 04JQ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 9078–9078. Navigation pointer invoked at proof line 10623; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original theorem 09YQ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7400–7400. Navigation pointer invoked at proof line 10567; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let ℓ be a prime number and n an integer > 0. Let K be a field with $G = Gal(K^{sep}/K)$ and let $H \subset G$ be a maximal pro-ℓ subgroup with L/K being the corresponding field extension. Then $H^n_{étale}(\mathrm{Spec}(K), \mathcal{F}) = 0$ for all ℓ-power torsion $\mathcal{F}$ if and only if $H^n_{étale}(\mathrm{Spec}(L), \underline{\mathbf{Z}/\ell\mathbf{Z}}) = 0$.*

Proof. Write $L = \bigcup L_i$ as the union of its finite subextensions over K. Our choice of H implies that $[L_i : K]$ is prime to ℓ. Thus $\mathrm{Spec}(L) = \lim_{i \in I} \mathrm{Spec}(L_i)$ as in Lemma 0DV7. Thus we may replace K by L and assume that the absolute Galois group G of K is a profinite pro-ℓ group.

Assume $H^n(\mathrm{Spec}(K), \underline{\mathbf{Z}/\ell\mathbf{Z}}) = 0$. Let $\mathcal{F}$ be an ℓ-power torsion sheaf on $\mathrm{Spec}(K)_{étale}$. We will show that $H^n_{étale}(\mathrm{Spec}(K), \mathcal{F}) = 0$. By the correspondence specified in Lemma 04JQ our sheaf $\mathcal{F}$ corresponds to an ℓ-power torsion G-module M. Any finite set of elements $x_1, \ldots, x_m \in M$ must be fixed by an open subgroup U by continuity. Let M' be the module spanned by the orbits of $x_1, \ldots, x_m$. This is a finite abelian ℓ-group as each x_i is killed by a power of ℓ and the orbits are finite. Since M is the filtered colimit of these submodules M', we see that $\mathcal{F}$ is the filtered colimit of the corresponding subsheaves $\mathcal{F}' \subset \mathcal{F}$. Applying Theorem 09YQ to this colimit, we reduce to the case where $\mathcal{F}$ is a finite locally constant sheaf.

Let M be a finite abelian ℓ-group with a continuous action of the profinite pro-ℓ group G. Then there is a G-invariant filtration

$$0 = M_0 \subset M_1 \subset \ldots \subset M_r = M$$

such that $M_{i+1}/M_i \cong \mathbf{Z}/\ell\mathbf{Z}$ with trivial G-action (this is a simple lemma on representation theory of finite groups; insert future reference here). Thus the corresponding sheaf $\mathcal{F}$ has a filtration

$$0 = \mathcal{F}_0 \subset \mathcal{F}_1 \subset \ldots \subset \mathcal{F}_r = \mathcal{F}$$

with successive quotients isomorphic to $\underline{\mathbf{Z}/\ell\mathbf{Z}}$. Thus by induction and the long exact cohomology sequence we conclude. □